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Introduction to Computer Architecture

Objectives:

- Introduction: Fundamentals of Computer Design, Introduction,
- RISC vs CISC, Computer Evolution and Performance.
- Fundamental of computer architecture, digital and analog.
- Digital computer system

1. What is Computer Design?

Computer design means how we **plan, organize, and build** a computer system — including its **hardware, software**, and how these parts work together to process data.

2. Basic Goals of Computer Design:

A well-designed computer system should be:

- **Fast** (high performance)
- **Low cost** (affordable)
- **Energy efficient** (less power usage)
- **Reliable** (no frequent crashes)

3. Key Elements in Computer Design:

- **Instruction Set Architecture (ISA):**
Defines the set of commands the CPU can understand.
- **Hardware Components:**
Includes CPU, memory, input/output devices, etc.
- **Performance Factors:**
Clock speed, number of cores, memory access time, etc.

4. Design Trade-offs:

Sometimes, we have to balance between:

- **Speed vs. Cost**
- **Performance vs. Power consumption**
- **Complexity vs. Simplicity**

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- Uses simple and few instructions.
- Each instruction is executed very fast (usually in one clock cycle).
- Easier to design and consumes less power.
- Commonly used in mobile phones and tablets (like ARM processors)

CISC (Complex Instruction Set Computer):

- Has many complex instructions that can do more work per instruction.
 - Instructions may take more than one clock cycle to complete.
 - More powerful for desktop and laptop computers (like Intel processors).
 - Makes programming easier by using fewer instructions to perform complex tasks.
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1. First Generation (1940s–1950s):

- Used **vacuum tubes**
- Very **large, slow**, and **high-power consumption**
- Example: ENIAC

2. Second Generation (1950s–1960s):

- Used **transistors** instead of vacuum tubes
- Smaller, **more reliable**, and **faster**

3. Third Generation (1960s–1970s):

- Used **integrated circuits (ICs)**
- Smaller, cheaper, and used **less power**

4. Fourth Generation (1970s–Present):

- Use of **microprocessors**
- High performance, **compact size**, and widely used in PCs

5. Fifth Generation (Now and Future):

- Based on **Artificial Intelligence, Cloud Computing, Quantum Computing**
 - Machines can now learn, make decisions, and work with huge data
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Computer Performance (How We Measure Speed):**1. Clock Speed (GHz):**

- Higher speed = more instructions per second
- Example: 3.0 GHz = 3 billion cycles per second

2. Number of Cores:

- Multi-core CPUs can perform **multiple tasks at once**
- Dual Core, Quad Core, Octa Core, etc.

3. RAM (Memory Size):

- More RAM = better performance in multitasking and large programs

4. Storage Type:

- **SSD (Solid State Drive)** is much faster than **HDD (Hard Disk Drive)**
- SSD helps in faster boot and load times

5. Architecture Design:

- Newer designs (like ARM, M1 chip) are more **efficient**
- Use less power and give more speed

6. Cache Memory & Bus Speed:

- Help in **faster data access** inside the CPU
- Improve performance for heavy tasks

NAME OF STUDENT: Muhammad Danish ID No: 470-2024**Digital Computers**

- Digital computers work using **binary numbers** (only 0 and 1).
- They process **discrete values**, which means fixed, step-by-step data.
- These computers perform calculations and logic operations very **accurately** and **quickly**.
- Digital systems are used in **almost all modern devices**, such as:
 - Personal computers (PCs)
 - Smartphones
 - Tablets
 - Laptops
 - Digital watches
- Digital computers are **easy to program**, **more reliable**, and **less affected by noise** or interference.

Analog Computers

- Analog computers work with **continuous values**, such as **voltage**, **pressure**, or **temperature**.
- Instead of binary, they process data that changes smoothly over time.
- Analog computers are used in **scientific, engineering, and measurement systems**, especially when dealing with physical quantities.
- Examples of analog devices include:
 - Old speedometers
 - Thermometers
 - Analog clocks
 - Weather instruments
- Analog computers are **faster** in some specific tasks but are generally **less accurate** and **more affected by noise**.

Main Differences Between Digital and Analog Computers

Feature	Digital Computer	Analog Computer
Data Type	Discrete (0 and 1)	Continuous (real-world values)
Accuracy	High accuracy	Less accurate
Usage	General-purpose (used everywhere)	Specific-purpose (scientific/engineering)
Speed	High speed for logical/calculations	Fast for continuous data
Common Today?	Yes, very common	Rare, used in special field

A Digital Computer System is a complete setup that processes data using digital signals (0s and 1s). It consists of several main components, each with a specific role to perform tasks efficiently.

1. Input Unit:

- This is the part of the computer that receives data and instructions from the user or external devices.
- It converts the input into digital form that the computer can understand.
- Examples of input devices are keyboards, mice, scanners, and microphones.
- Without input, the computer cannot perform any operation.

NAME OF STUDENT: Muhammad Danish ID No: 470-2024**2. Memory Unit:**

- The memory stores data and instructions needed for processing.
 - There are two main types of memory:
 - **Primary Memory (Main Memory):** This includes RAM (Random Access Memory), which temporarily holds data and instructions during processing. It is fast but volatile (loses data when power is off).
 - **Secondary Memory:** Includes hard drives, SSDs, and other storage devices that store data permanently.
 - Memory plays a crucial role in the speed and performance of the computer.
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3. Central Processing Unit (CPU):

- The CPU is called the **brain** of the computer because it controls all activities and processes all data.
 - It has two important parts:
 - **Control Unit (CU):** Directs the flow of data between the CPU, memory, and input/output devices. It interprets instructions and tells the computer what to do.
 - **Arithmetic Logic Unit (ALU):** Performs all mathematical calculations (addition, subtraction, etc.) and logical operations (comparing values, decision making).
 - The CPU executes instructions step-by-step to complete tasks.
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4. Output Unit:

- This unit takes the processed data from the CPU and presents it to the user in a usable form.
- Output devices include monitors, printers, speakers, and projectors.
- It converts digital data back into human-readable or usable form.

Lab no 1 Tasks

1. Identify and Explain Basic Components of a Computer

List the main parts of a digital computer system (Input unit, CPU, Memory, Output unit). Describe the function of each part in your own words.

Input Unit:

Receives data and commands from external sources (such as a user typing or clicking) and translates them into a digital format the machine can process.

Central Processing Unit (CPU):

Acts as the command center, executing software instructions, performing mathematical and logical calculations, and directing the flow of information between all other hardware components.

Memory Unit:

Stores data and processing instructions. It is typically divided into fast, temporary primary memory (RAM) for active tasks and permanent secondary memory (hard drives or SSDs) for long-term storage.

NAME OF STUDENT: Muhammad Danish ID No: 470-2024**Output Unit:**

Converts the computer's processed digital data into human-readable or usable formats, such as displaying visuals on a screen, printing documents, or playing audio through speakers.

Communication Unit:

Handles the transmission and reception of data across networks. It uses hardware like network interface cards (NICs) or Wi-Fi adapters to connect the computer to the internet, local networks, or other external devices.

2. Compare RISC and CISC Architectures

Make a table listing at least 3 differences between RISC and CISC.

Give examples of devices or processors that use each type.

Feature	RISC	CISC
Instructions	Simple; one operation per instruction.	Complex; multiple operations per instruction.
Execution Speed	Fast; usually takes one clock cycle.	Slower; often takes multiple clock cycles.
Power & Design	Simpler hardware, highly energy-efficient.	Complex hardware, higher power consumption.
Examples	ARM processors (smartphones, Apple M-chips).	Intel and AMD processors (desktop PCs, laptops).

NAME OF STUDENT: Muhammad Danish ID No: 470-2024**3. Analyze Computer Performance Parameters**

Explain what factors affect computer performance (clock speed, cores, RAM, storage). Find the specifications of two different computers or smartphones and compare their performance based on these factors.

Performance Factors:

- **Clock Speed (GHz):**
Higher speed means faster processing of instructions.
- **Number of Cores:**
More cores allow the CPU to perform multiple tasks simultaneously.
- **RAM:**
More memory provides smoother multitasking and handles larger programs better.
- **Storage Type:**
Solid State Drives (SSDs) offer significantly faster boot and app loading times compared to HDDs.

Comparison: Samsung Galaxy S24 vs. Apple iPhone 15

Feature	Samsung Galaxy S24	Apple iPhone 15
Cores	10 Cores	6 Cores
Clock Speed	Up to 3.20 GHz	Up to 3.46 GHz
RAM	8 GB	6 GB
Storage	UFS 4.0 (Solid State)	NVMe (Solid State)

Analysis:

The iPhone 15 has a higher peak clock speed, giving it an advantage in fast, single-core tasks. The Galaxy S24 features more cores and RAM, making it better suited for heavy, background multitasking. Both use highly efficient solid-state storage for instant load-time.